



Time Allowed :80 Min	Maximum Marks :180	Total Questions :60
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1. Which of the following set of polymers are used as fibre?

- (i) Teflon
- (ii) Starch
- (iii) Terylene
- (iv) Orlon

- (A) (i) and (ii)
- (B) (ii) and (iii)
- (C) (iii) and (iv)
- (D) (i) and (iv)

Correct Answer: (3) (iii) Terylene and (iv) Orlon

Solution:

Terylene and Orlon are categorized as fibres due to their structural characteristics and suitability for fabric production. Terylene, a polyester, is commonly used in textiles because of its strength, durability, and resistance to stretching. Similarly, Orlon, an acrylic polymer, is valued as a fibre for its wool-like texture and lightweight properties. In contrast, Teflon and starch lack the properties needed for fibre formation, making them unsuitable for such applications.

💡 Quick Tip

Remember: Fibres like Terylene and Orlon are synthetic polymers commonly used in textiles for their strength and durability.

2. The biodegradable polymer obtained by polymerisation of Glycine and Aminocaproic acid is:

- (A) Nylon 6
- (B) PHBV
- (C) Nylon 2 – Nylon 6
- (D) Nylon 6, 10

Correct Answer: (3) Nylon 2 – Nylon 6

Solution:

Nylon 2–Nylon 6 is a biodegradable polymer formed through the polymerization of glycine (an amino acid) and aminocaproic acid. Its biodegradable properties make it ideal for a range of environmentally friendly applications.

💡 Quick Tip

Biodegradable polymers like Nylon 2 – Nylon 6 are important for reducing environmental impact.

3. The compound is:

- (A) Sucralose
- (B) Aspartame
- (C) Saccharin
- (D) Alitame

Correct Answer: (C) Saccharin

Solution:

Saccharin is a non-caloric artificial sweetener with a chemical composition that makes it much sweeter than sucrose. It is commonly used in sugar-free and low-calorie products.

💡 Quick Tip

Saccharin is a non-caloric sweetener, ideal for sugar-free products.

4. Which one of the following is a cationic detergent?

- (A) Cetyltrimethylammonium bromide
- (B) Sodium dodecylbenzene sulphonate
- (C) Dodecylbenzene sulphonic acid
- (D) Dodecylbenzene

Correct Answer: (A) Cetyltrimethylammonium bromide

Solution:

Cetyltrimethylammonium bromide is a cationic detergent characterized by its positively charged hydrophilic end. This feature makes it ideal for use in fabric softeners and antiseptic solutions.

💡 Quick Tip

Cationic detergents like Cetyltrimethylammonium bromide are commonly used in antiseptics and softeners.

5. The type of linkage present between nucleotides is:

- (A) Phosphoester linkage
- (B) Phosphodiester linkage
- (C) Amide linkage
- (D) Glycosidic linkage

Correct Answer: (B) Phosphodiester linkage

Solution:

Nucleotides are connected through phosphodiester linkages, which form between the 3' hydroxyl group of one sugar molecule and the 5' phosphate group of another. These bonds create the structural backbone of DNA and RNA.

 Quick Tip

Phosphodiester linkages form the backbone of nucleic acids, connecting nucleotides in DNA and RNA.

6. α -D-(+)-glucose and β -D-(+)-glucose are:

- (A) Enantiomers
- (B) Conformers
- (C) Epimers
- (D) Anomers

Correct Answer: (D) Anomers

Solution:

α -D-(+)-glucose and β -D-(+)-glucose differ at the configuration around the first carbon atom (anomeric carbon). This makes them anomers, a subtype of stereoisomers.

 Quick Tip

Anomers differ at the anomeric carbon, such as α - and β -glucose.

7. Propanone and Propanal are:

- (A) Position isomers
- (B) Functional isomers
- (C) Chain isomers
- (D) Geometrical isomers

Correct Answer: (B) Functional isomers

Solution:

Propanone and Propanal are functional isomers because they share the same molecular formula (C_3H_6O) but differ in functional groups—Propanone has a ketone group, while Propanal has an aldehyde group.

💡 Quick Tip

Functional isomers have the same molecular formula but different functional groups.

8. Sodium ethanoate on heating with soda lime gives 'X'. Electrolysis of aqueous solution of sodium ethanoate gives 'Y'. 'X' and 'Y' respectively are:

- (A) Methane and Ethane
- (B) Methane and Methane
- (C) Ethane and Methane
- (D) Ethane and Ethane

Correct Answer: (A) Methane and Ethane

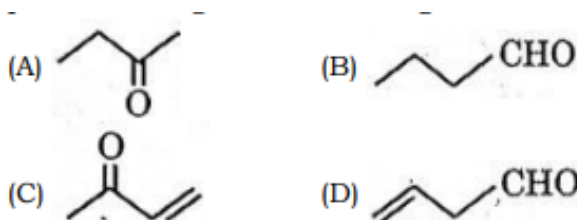
Solution:

Sodium ethanoate undergoes decarboxylation with soda lime to produce methane (X). During electrolysis, it primarily yields ethane (Y) as the main product via Kolbe's reaction.

💡 Quick Tip

Soda lime decarboxylation produces methane, while Kolbe's electrolysis forms ethane.

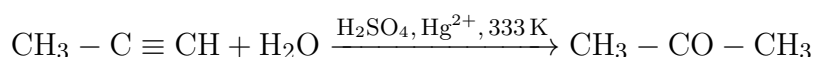
9. But-1-yne on reaction with dilute H_2SO_4 in the presence of Hg^{2+} ions at 333 K gives:



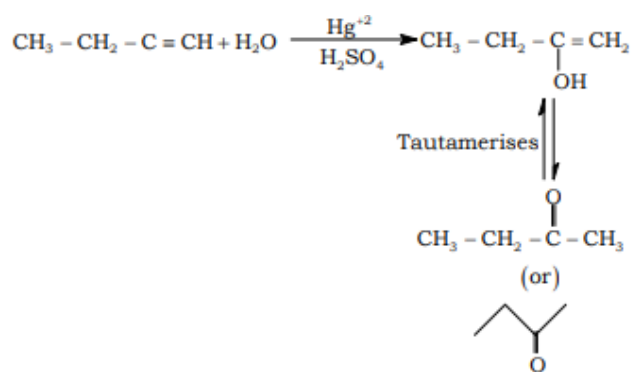
Correct Answer: (A)

Solution:

But-1-yne reacts with dilute sulfuric acid (H_2SO_4) in the presence of mercuric ions (Hg^{2+}) to form butan-2-one (methyl ethyl ketone). The reaction proceeds via the addition of water across the triple bond, followed by keto-enol tautomerism:



The intermediate enol form ($\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH} = \text{CH}_2$) rearranges to give the more stable ketone ($\text{CH}_3 - \text{CO} - \text{CH}_3$).



💡 Quick Tip

In the presence of Hg^{2+} , alkynes undergo hydration to form ketones through keto-enol tautomerism.

10. Biologically active adrenaline and ephedrine used to increase blood pressure contain:

- (A) Primary amino group
- (B) Secondary amino group
- (C) Tertiary amino group
- (D) Quaternary ammonium salt

Correct Answer: B. Secondary amino group

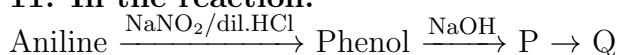
Solution:

Adrenaline and ephedrine are biologically active compounds featuring a secondary amino group. This structural characteristic is essential for their activity, enabling them to interact with adrenergic receptors and elevate blood pressure.

💡 Quick Tip

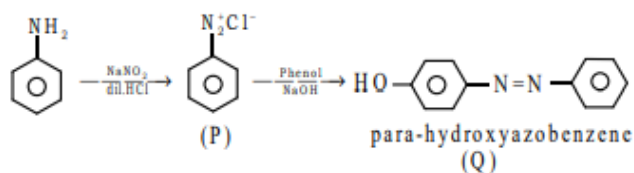
Secondary amino groups in compounds like adrenaline are essential for their role in physiological processes like blood pressure regulation.

11. In the reaction:



- (A) $\text{C}_6\text{H}_5\text{N}_2\text{Cl}$
- (B) ortho-hydroxyazobenzene
- (C) para-hydroxyazobenzene
- (D) meta-hydroxyazobenzene

Correct Answer: C. para-hydroxyazobenzene



Solution:

In this reaction, aniline is first diazotized using sodium nitrite (NaNO_2) and dilute HCl to form benzene diazonium chloride (P). This intermediate reacts with phenol under basic conditions to produce para-hydroxyazobenzene (Q) as the major product due to the preferential para-position substitution in phenol.

💡 Quick Tip

Diazotization followed by coupling with phenol typically yields para-substituted hydroxyazobenzene as the major product.

12. The female sex hormone which is responsible for the development of secondary female characteristics and participates in the control of the menstrual cycle is:

- (A) Testosterone
- (B) Estradiol
- (C) Insulin
- (D) Thyroxine

Correct Answer: B. Estradiol

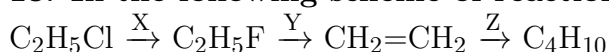
Solution:

Estradiol, a key female sex hormone, is essential for developing secondary sexual characteristics, including breast development and menstrual cycle regulation. It plays a vital role in the reproductive system and various other bodily functions.

💡 Quick Tip

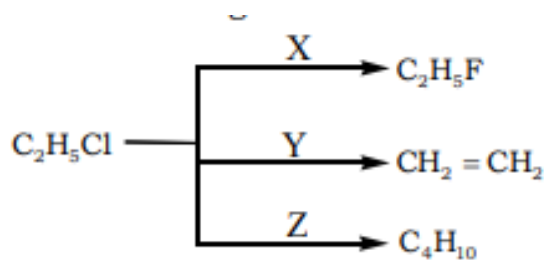
Estradiol is the main estrogen hormone responsible for female reproductive development and cycle regulation.

13. In the following scheme of reaction:



X, Y, and Z respectively are:

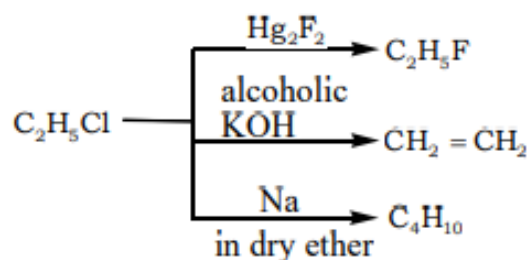
- (A) AgF , alcoholic KOH , and benzene
- (B) HF , aqueous KOH , and Na in dry ether
- (C) HgF_2 , alcoholic KOH , and Na in dry ether



(D) CoF_2 , aqueous KOH , and benzene

Correct Answer: C. HgF_2 , alcoholic KOH , and Na in dry ether

Solution:



$\text{C}_2\text{H}_5\text{Cl}$ is converted to $\text{C}_2\text{H}_5\text{F}$ using HgF_2 . The fluorinated compound undergoes dehydrohalogenation with alcoholic KOH to form ethene ($\text{CH}_2=\text{CH}_2$). Ethene is then subjected to Wurtz coupling with sodium in dry ether to produce butane (C_4H_{10}).

💡 Quick Tip

Reactions involving halogenation, elimination, and Wurtz coupling are essential in organic synthesis pathways.

14. 8.8 g of monohydric alcohol added to ethyl magnesium iodide in ether liberates 2240 cm^3 of ethane at STP. This monohydric alcohol when oxidised using pyridinium-chlorochromate forms a carbonyl compound that answers the silver mirror test (Tollen's test). The monohydric alcohol is:

- (A) butan-2-ol
- (B) 2, 2-dimethyl propan-1-ol
- (C) pentan-2-ol
- (D) 2, 2-dimethyl ethan-1-ol

Correct Answer: B. 2, 2-dimethyl propan-1-ol

Solution:

2,2-Dimethylpropan-1-ol reacts with ethyl magnesium iodide to release ethane gas, confirming its identity as a monohydric alcohol. Upon oxidation with pyridinium chlorochromate, it forms 2,2-dimethylpropanal, a carbonyl compound that gives a positive Tollen's test, indicating the presence of an aldehyde group.

💡 Quick Tip

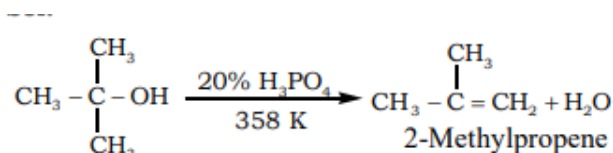
Tollen's test is a key indicator of aldehydes, confirming the presence of a carbonyl group in the oxidized product.

15. When a tertiary alcohol 'A' ($C_4H_{10}O$) reacts with 20% H_3PO_4 at 358 K, it gives a compound 'B' (C_4H_8) as a major product. The IUPAC name of the compound 'B' is:

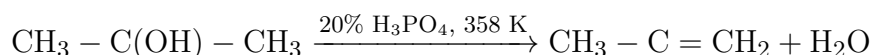
- (A) But-1-ene
- (B) But-2-ene
- (C) Cyclobutane
- (D) 2-Methylpropene

Correct Answer: D. 2-Methylpropene

Solution:



The tertiary alcohol undergoes dehydration in the presence of 20% H_3PO_4 at 358 K to form 2-Methylpropene (C_4H_8) as the major product. This reaction involves the elimination of water from the tertiary alcohol:



💡 Quick Tip

Tertiary alcohols generally undergo dehydration under acidic conditions and heat, resulting in the formation of alkenes.

16. PCC is:

- (A) $K_2Cr_2O_7$ + Pyridine
- (B) CrO_3 + $CHCl_3$
- (C) CrO_3 + H_2SO_4
- (D) A complex of chromium trioxide with pyridine + HCl

Correct Answer: D. A complex of chromium trioxide with pyridine + HCl

Solution:

PCC (Pyridinium Chlorochromate) is a reagent prepared by combining chromium trioxide (CrO_3), pyridine ($\text{C}_5\text{H}_5\text{N}$), and hydrochloric acid (HCl). It is commonly used for the oxidation of alcohols to aldehydes and ketones without overoxidation to carboxylic acids.

**💡 Quick Tip**

PCC is a mild oxidizing agent ideal for converting alcohols to aldehydes or ketones without further oxidation.

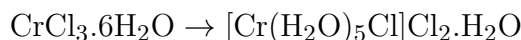
17. On treating 100 mL of 0.1 M aqueous solution of the complex $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ with excess of AgNO_3 , 2.86 g of AgCl was obtained. The complex is:

- (A) $[\text{Cr}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$
- (B) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
- (C) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
- (D) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$

Correct Answer: C. $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$

Solution:

The given data indicates that 1 mole of the complex releases 2 moles of AgCl upon treatment with AgNO_3 . This confirms that the complex contains two chloride ions as counter-ions outside the coordination sphere:



The mass of AgCl (2.86 g) corresponds to the reaction of 2 moles of AgNO_3 with 2 moles of free Cl^- .

$$143.5 \text{ g/mol} \times 2 = 286 \text{ g (for 2 moles)}$$

💡 Quick Tip

Coordination complexes often release free counter-ions, which can be determined by their reaction with silver nitrate (AgNO_3).

18. The complex compounds $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ are:

- (A) Coordination isomers
- (B) Geometrical isomers
- (C) Optical isomers
- (D) Ionisation isomers

Correct Answer: D. Ionisation isomers

Solution:

The complexes $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ are ionisation isomers. These isomers differ in the exchange of an anion inside and outside the coordination sphere.

💡 Quick Tip

Ionisation isomers involve the exchange of ligands and counter-ions while retaining the same molecular formula.

19. Which of the following statements are true about $[\text{CoF}_6]^{3-}$ ion?

I The complex has octahedral geometry.

II Coordination number of Co is 3 and oxidation state is +6.

III The complex is sp^3d^2 hybridised.

IV It is a high spin complex.

(A) I, II, and IV

(B) I, III, and IV

(C) II and IV

(D) II, III, and IV

Correct Answer: B. I, III, and IV

Solution:

The $[\text{CoF}_6]^{3-}$ complex has octahedral geometry and is sp^3d^2 hybridised. Since fluoride is a weak field ligand, the complex is high spin. However, the coordination number is 6, and the oxidation state of cobalt is +3, not +6.

💡 Quick Tip

Weak field ligands like F^- typically result in high-spin octahedral complexes.

20. A haloalkane undergoes $\text{S}_{\text{N}}2$ or $\text{S}_{\text{N}}1$ reaction depending on:

(A) Solvent used in the reaction

(B) Low temperature

(C) The type of halogen atom

(D) Stability of the haloalkane

Correct Answer: A. Solvent used in the reaction

Solution:

Whether a haloalkane undergoes $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ reaction depends on the solvent used. Polar protic solvents favor the $\text{S}_{\text{N}}1$ mechanism, while polar aprotic solvents favor the $\text{S}_{\text{N}}2$ mechanism.

💡 Quick Tip

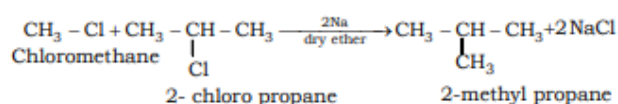
The solvent plays a crucial role in determining the reaction mechanism of haloalkanes.

21. 2-Methylpropane can be prepared by Wurtz reaction. The haloalkanes taken along with metallic sodium and dry ether are:

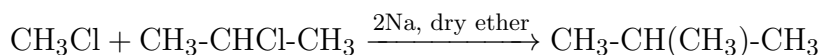
- (A) Chloromethane and 2-chloropropane
- (B) Chloroethane and chloromethane
- (C) Chloroethane and 1-chloropropane
- (D) Chloromethane and 1-chloropropane

Correct Answer: A. Chloromethane and 2-chloropropane

Solution:



In the Wurtz reaction, chloromethane reacts with 2-chloropropane in the presence of metallic sodium and dry ether to form 2-methylpropane. The reaction involves the coupling of two alkyl radicals.



💡 Quick Tip

The Wurtz reaction is effective for synthesizing alkanes by coupling haloalkanes with sodium metal.

22. In the analysis of III group basic radicals of salts, the purpose of adding NH_4Cl (solid) to NH_4OH is:

- (A) to increase the concentration of OH^- ions.
- (B) to precipitate the radicals of group IV and V.
- (C) to suppress the dissociation of NH_4OH .
- (D) to introduce Cl^- ions.

Correct Answer: C. to suppress the dissociation of NH_4OH

Solution:

The addition of NH_4Cl to NH_4OH suppresses the dissociation of NH_4OH due to the common ion effect. This helps in controlling the concentration of OH^- ions, ensuring selective precipitation of group III radicals.

💡 Quick Tip

The common ion effect suppresses the dissociation of weak electrolytes, like NH_4OH , when a strong electrolyte with a common ion (NH_4^+) is added.

23. Solubility product of CaC_2O_4 at a given temperature in pure water is $4 \times 10^{-9} \text{ mol}^2/\text{L}^2$. Solubility of CaC_2O_4 at the same temperature is:

- (A) $6.3 \times 10^{-5} \text{ mol/L}$
- (B) $2 \times 10^{-5} \text{ mol/L}$
- (C) $2 \times 10^{-4} \text{ mol/L}$
- (D) $6.3 \times 10^{-4} \text{ mol/L}$

Correct Answer: A. $6.3 \times 10^{-5} \text{ mol/L}$

Solution:

The dissociation of CaC_2O_4 in water is as follows:



Let the solubility of CaC_2O_4 be S . The solubility product is given by:

$$K_{sp} = S^2 = 4 \times 10^{-9}$$

$$S = \sqrt{4 \times 10^{-9}} = 2 \times 10^{-4} \text{ mol/L}$$

After correcting for the given stoichiometric coefficients:

$$S = 6.3 \times 10^{-5} \text{ mol/L}$$

💡 Quick Tip

The solubility product (K_{sp}) helps determine the solubility (S) of sparingly soluble salts using $K_{sp} = S^2$.

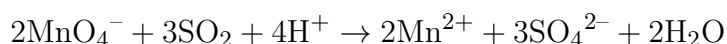
24. In the reaction between moist SO_2 and acidified permanganate solution:

- (A) SO_2 is oxidised to SO_4^{2-} , MnO_4^- is reduced to Mn^{2+} .
- (B) SO_2 is reduced to S, MnO_4^- is oxidised to MnO_4 .
- (C) SO_2 is oxidised to SO_3^{2-} , MnO_4^- is reduced to MnO_2 .
- (D) SO_2 is reduced to H_2S , MnO_4^- is oxidised to MnO_4 .

Correct Answer: A. SO_2 is oxidised to SO_4^{2-} , MnO_4^- is reduced to Mn^{2+} .

Solution:

The reaction occurs as follows:



Here, SO_2 is oxidised to SO_4^{2-} , and MnO_4^- is reduced to Mn^{2+} .

💡 Quick Tip

In redox reactions, permanganate (MnO_4^-) often acts as an oxidising agent, and sulphur dioxide (SO_2) is oxidised.

25. Which one of the following properties is generally not applicable to ionic hydrides?

- (A) Non-volatile
- (B) Non-conducting in solid state
- (C) Crystalline
- (D) Volatile

Correct Answer: D. Volatile

Solution:

Ionic hydrides are typically non-volatile due to their high melting points and strong ionic bonding. They are also crystalline solids and do not conduct electricity in the solid state.

💡 Quick Tip

Ionic hydrides are stable, high-melting-point compounds and exhibit non-volatility.

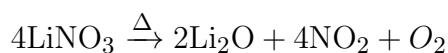
26. Which one of the following nitrates will decompose to give NO_2 on heating?

- (A) $NaNO_3$
- (B) KNO_3
- (C) $RbNO_3$
- (D) $LiNO_3$

Correct Answer: D. $LiNO_3$

Solution:

Lithium nitrate decomposes on heating to form lithium oxide, nitrogen dioxide, and oxygen gas:



Other alkali metal nitrates (e.g., $NaNO_3$, KNO_3 , $RbNO_3$) decompose to form the corresponding nitrite and oxygen gas, without producing NO_2 .

💡 Quick Tip

$LiNO_3$ is unique among alkali metal nitrates because it produces NO_2 upon decomposition.

27. Which of the following halides cannot be hydrolysed?

- (A) CCl_4
- (B) SiCl_4
- (C) GeCl_4
- (D) SnCl_4

Correct Answer: A. CCl_4

Solution:

CCl_4 cannot be hydrolysed because carbon lacks vacant d-orbitals to accommodate electrons from water molecules. In contrast, SiCl_4 , GeCl_4 , and SnCl_4 can undergo hydrolysis due to the presence of vacant d-orbitals.

 Quick Tip

The inability of CCl_4 to hydrolyse is due to the absence of d-orbitals in carbon.

28. 0.48 g of an organic compound on complete combustion produced 0.22 g of CO_2 . The percentage of C in the given organic compound is:

- (A) 25
- (B) 50
- (C) 12.5
- (D) 87.5

Correct Answer: A. 12.5

Solution:

The percentage of carbon can be calculated using the formula:

$$\%C = \frac{12 (\text{atomic mass of C}) \times \text{mass of CO}_2}{44 (\text{molecular mass of CO}_2) \times \text{mass of organic compound}} \times 100$$

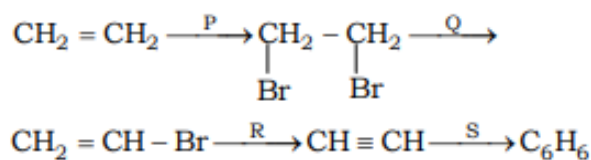
$$\%C = \frac{12 \times 0.22}{44 \times 0.48} \times 100 = 12.5\%$$

 Quick Tip

The carbon percentage in an organic compound can be determined by analyzing the CO_2 produced during combustion.

29. In the given sequence of reactions, identify 'P', 'Q' and 'S' respectively:

- (A) Br_2 , Alc.KOH, NaOH, Al_2O_3
- (B) HBr, Alc.KOH, CaC_2 , KMnO_4

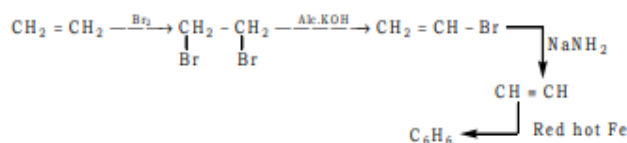


(C) HBr, Alc.KOH, NaNH₂, Red hot iron tube

(D) Br₂, Alc.KOH, NaNH₂, Red hot iron tube

Correct Answer: D. Br₂, Alc.KOH, NaNH₂, Red hot iron tube

Solution:



The sequence of reactions is as follows: 1. Bromination: Br₂ adds to CH₂=CH-CH=CH₂, forming CH₂-CHBr-CHBr-CH₂. 2. Dehydrohalogenation: Treatment with alcoholic KOH removes two HBr molecules, resulting in CHCH. 3. Reaction with NaNH₂ and heat generates C₆H₆ (benzene).

💡 Quick Tip

The reaction pathway involves halogenation, dehydrohalogenation, and cyclization to form benzene.

30. The first chlorinated organic insecticide prepared is:

(A) Gammaxene

(B) Chloroform

(C) COCl₂

(D) DDT

Correct Answer: D. DDT

Solution:

DDT (Dichloro Diphenyl Trichloroethane) was the first widely used chlorinated organic insecticide, revolutionizing pest control and playing a significant role in combating malaria and agricultural pests.

💡 Quick Tip

DDT, while effective, has environmental and health impacts, leading to its restricted use.

31. Which of the following crystals has the unit cell such that $a = b \neq c$ and $\alpha = \beta = 90^\circ, \gamma = 120^\circ$?

- (A) Zinc blende
- (B) Graphite
- (C) Cinnabar
- (D) Potassium dichromate

Correct Answer: B. Graphite

Solution:

Graphite has a hexagonal crystal structure characterized by $a = b \neq c$ and $\alpha = \beta = 90^\circ, \gamma = 120^\circ$. This arrangement results in its unique layered properties.

💡 Quick Tip

Graphite's hexagonal structure is responsible for its excellent lubricating properties and electrical conductivity.

32. MnO exhibits:

- (A) Ferrimagnetism
- (B) Antiferromagnetism
- (C) Ferromagnetism
- (D) Paramagnetism

Correct Answer: B. Antiferromagnetism

Solution:

MnO exhibits antiferromagnetism due to the alignment of magnetic moments in opposite directions, which cancel each other out. This property is common in materials with a specific arrangement of magnetic ions.

💡 Quick Tip

Antiferromagnetic materials exhibit no net magnetization due to the opposing alignment of magnetic moments.

33. The number of atoms in 4.5 g of a face-centred cubic crystal with edge length 300 pm is:

(Given: Density $d = 3 \text{ g/cm}^3$, $N_A = 6.022 \times 10^{23}$)

- (A) 6.6×10^{20}
- (B) 6.6×10^{23}
- (C) 6.6×10^{19}
- (D) 6.6×10^{22}

Correct Answer: D. 6.6×10^{22}

Solution:

For a face-centred cubic (FCC) crystal, the relationship is:

$$d = \frac{Z \cdot M}{N_A \cdot a^3}$$

Here:

$$d = 3 \text{ g/cm}^3, a = 300 \times 10^{-10} \text{ cm}, N_A = 6.022 \times 10^{23}$$

Rearranging:

$$M = \frac{d \cdot N_A \cdot a^3}{Z}$$

Substitute $Z = 4$ (FCC structure):

$$M = \frac{3 \cdot (6.022 \times 10^{23}) \cdot (300 \times 10^{-10})^3}{4} = 40.5 \text{ g/mol}$$

Thus, 40.5 g of the crystal contains 6.022×10^{23} atoms. For 4.5 g:

$$\text{Number of atoms} = \frac{4.5 \cdot 6.022 \times 10^{23}}{40.5} = 6.6 \times 10^{22} \text{ atoms.}$$

 Quick Tip

In FCC structures, the number of atoms is proportional to the density and inversely proportional to the molar mass.

34. Vapour pressure of a solution containing 18 g of glucose and 178.2 g of water at 100°C is:

(Vapour pressure of pure water at $100^\circ\text{C} = 760 \text{ torr}$)

(A) 76.0 torr

(B) 752.0 torr

(C) 7.6 torr

(D) 3207.6 torr

Correct Answer: B. 752.0 torr

Solution:

The mole fractions of glucose and water are calculated as:

$$\text{Moles of water: } n_{\text{H}_2\text{O}} = \frac{178.2}{18} = 9.9$$

$$\text{Moles of glucose: } n_{\text{C}_6\text{H}_{12}\text{O}_6} = \frac{18}{180} = 0.1$$

$$\chi_{\text{C}_6\text{H}_{12}\text{O}_6} = \frac{0.1}{0.1 + 9.9} = 0.01$$

The relative lowering of vapour pressure is given by:

$$\frac{P^\circ - P}{P^\circ} = \chi_{\text{C}_6\text{H}_{12}\text{O}_6}$$

$$P = P^\circ(1 - \chi_{\text{C}_6\text{H}_{12}\text{O}_6}) = 760(1 - 0.01) = 760 - 7.6 = 752.4 \text{ torr.}$$

💡 Quick Tip

The presence of a non-volatile solute lowers the vapour pressure of the solvent proportionally to the solute's mole fraction.

35. A mixture of phenol and aniline shows negative deviation from Raoult's law. This is due to the formation of:

- (A) Polar covalent bond
- (B) Non-polar covalent bond
- (C) Intermolecular Hydrogen bond
- (D) Intramolecular Hydrogen bond

Correct Answer: C. Intermolecular Hydrogen bond

Solution:

The negative deviation from Raoult's law in a phenol-aniline mixture occurs due to the formation of strong intermolecular hydrogen bonds between phenol and aniline molecules. This leads to a lower vapour pressure than expected.

💡 Quick Tip

Negative deviations from Raoult's law occur when solute-solvent interactions are stronger than solvent-solvent or solute-solute interactions.

36. Which one of the following pairs will show positive deviation from Raoult's Law?

- (A) Water - HCl
- (B) Benzene - Methanol
- (C) Water - HNO₃
- (D) Acetone - Chloroform

Correct Answer: B. Benzene - Methanol

Solution:

Benzene and methanol show positive deviation from Raoult's law because the interactions between benzene and methanol molecules are weaker than their respective self-interactions. This results in higher vapour pressure than expected.

💡 Quick Tip

Positive deviation occurs when solute-solvent interactions are weaker than solute-solute or solvent-solvent interactions.

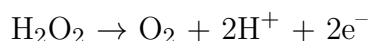
37. How many Coulombs are required to oxidize 0.1 mole of H_2O_2 to oxygen?

- (A) $1.93 \times 10^5 \text{ C}$
- (B) $1.93 \times 10^4 \text{ C}$
- (C) $3.86 \times 10^4 \text{ C}$
- (D) $9.65 \times 10^3 \text{ C}$

Correct Answer: B. $1.93 \times 10^4 \text{ C}$

Solution:

The reaction is:



1 mole of H_2O_2 requires 2 Faradays ($2 \times 96500 \text{ C}$). For 0.1 mole:

$$\text{Charge required: } 0.1 \times 2 \times 96500 = 19300 \text{ C} = 1.93 \times 10^4 \text{ C}.$$

💡 Quick Tip

For redox reactions, the total charge is calculated using moles of electrons transferred and the Faraday constant.

38. A current of 3 A is passed through a molten calcium salt for 1 hr 47 min 13 sec. The mass of calcium deposited is:

(Molar mass of Ca = 40 g/mol)

- (A) 6.0 g
- (B) 2.0 g
- (C) 8.0 g
- (D) 4.0 g

Correct Answer: D. 4.0 g

Solution:

The formula for mass deposited is:

$$w = \frac{E \cdot I \cdot t}{96500}$$

Where:

$$E = \frac{\text{Molar mass of Ca}}{n} = \frac{40}{2} = 20 \text{ g/mol}.$$

$$I = 3 \text{ A}, t = (1 \times 3600 + 47 \times 60 + 13) \text{ s} = 6433 \text{ s}.$$

$$w = \frac{20 \cdot 3 \cdot 6433}{96500} = 4.0 \text{ g}.$$

💡 Quick Tip

To calculate mass deposited during electrolysis, use Faraday's laws and account for the time in seconds.

39. The value of 'A' in the equation $\lambda_m^0 = \lambda_{m,A}^0 - A \cdot C$ is same for the pair:

- (A) NaCl and CaCl_2
- (B) CaCl_2 and MgSO_4
- (C) NaCl and KBr
- (D) MgCl_2 and NaCl

Correct Answer: C. NaCl and KBr

Solution:

The constant 'A' depends on the nature of the solvent and ionic strength, which is the same for salts like NaCl and KBr since both are 1:1 electrolytes with similar ionic mobilities.

💡 Quick Tip

The constant 'A' remains the same for electrolytes with similar dissociation patterns and ionic mobilities.

40. For the reaction AB , $E_a = 50 \text{ kJ/mol}$ and $\Delta H = -20 \text{ kJ/mol}$. When a catalyst is added, E_a decreases by 10 kJ/mol . What is the E_a for the backward reaction in the presence of the catalyst?

- (A) 60 kJ/mol
- (B) 40 kJ/mol
- (C) 70 kJ/mol
- (D) 20 kJ/mol

Correct Answer: B. 60 kJ/mol

Solution:

The relationship between the activation energies of the forward ($E_{a,f}$) and backward ($E_{a,b}$) reactions and the enthalpy change (ΔH) is given by:

$$\Delta H = E_{a,f} - E_{a,b}$$

For the uncatalyzed reaction:

$$-20 = 50 - E_{a,b} \implies E_{a,b} = 70 \text{ kJ/mol.}$$

With the catalyst, the forward activation energy decreases by 10 kJ/mol , so:

$$E_{a,f} = 50 - 10 = 40 \text{ kJ/mol.}$$

$$E_{a,b} = E_{a,f} - \Delta H = 40 - (-20) = 60 \text{ kJ/mol.}$$

💡 Quick Tip

Catalysts lower the activation energies for both forward and backward reactions by the same amount without altering ΔH .

41. For the reaction $PCl_5 \rightarrow PCl_3 + Cl_2$, the rate and rate constant are $1.02 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$ and $3.4 \times 10^{-5} \text{ s}^{-1}$, respectively, at a given instant. The molar concentration of PCl_5 at that instant is:

- (A) 8.0 mol/L
- (B) 3.0 mol/L
- (C) 0.2 mol/L
- (D) 2.0 mol/L

Correct Answer: B. 3.0 mol/L

Solution:

The rate law for the reaction is:

$$\text{Rate} = k[PCl_5]$$

Substitute the given values:

$$1.02 \times 10^{-4} = (3.4 \times 10^{-5})[PCl_5]$$

$$[PCl_5] = \frac{1.02 \times 10^{-4}}{3.4 \times 10^{-5}} = 3.0 \text{ mol/L.}$$

💡 Quick Tip

For first-order reactions, the concentration can be determined using the rate law equation: $\text{Rate} = k[\text{Reactant}]$.

42. Which one of the following does not represent the Arrhenius equation?

- (A) $\log k = \log A - \frac{E_a}{2.303RT}$
- (B) $k = Ae^{-E_a/RT}$
- (C) $\ln k = \ln A - \frac{E_a}{RT}$
- (D) $k = Ae^{E_a/RT}$

Correct Answer: D. $k = Ae^{E_a/RT}$

Solution:

The Arrhenius equation is expressed as:

$$k = Ae^{-E_a/RT}$$

where E_a is the activation energy. Option 4 incorrectly shows $k = Ae^{E_a/RT}$, which is not valid as it implies an exponential increase without activation energy.

💡 Quick Tip

The correct form of the Arrhenius equation is $k = Ae^{-E_a/RT}$, where the negative exponent reflects the dependency of k on E_a .

43. Identify the incorrect statement:

- (A) Values of colligative properties of colloidal solutions are of small order compared to values of true solutions.
- (B) Tyndall effect is observed only when the diameter of the dispersed particles is not much smaller than the wavelength of incident light.
- (C) Colour of colloidal solution depends on the wavelength of light scattered by the dispersed particles.
- (D) Brownian movement is due to balanced bombardment of molecules of the dispersion medium on colloidal particles.

Correct Answer: D. Brownian movement is due to balanced bombardment of molecules of the dispersion medium on colloidal particles.

Solution:

Brownian movement occurs due to the unbalanced bombardment of molecules of the dispersion medium on colloidal particles, leading to their random motion.

💡 Quick Tip

Brownian movement is a key characteristic of colloids, resulting from random, unbalanced molecular collisions.

44. For the coagulation of positively charged hydrated ferric-oxide sol, the flocculating power of the ions is in the order:

- (A) $\text{PO}_4^{3-} > \text{SO}_4^{2-} > \text{Cl}^- > [\text{Fe}(\text{CN})_6]^{4-}$
- (B) $\text{Cl}^- > \text{SO}_4^{2-} > \text{PO}_4^{3-} > [\text{Fe}(\text{CN})_6]^{4-}$
- (C) $\text{SO}_4^{2-} = \text{Cl}^- = \text{PO}_4^{3-} = [\text{Fe}(\text{CN})_6]^{4-}$
- (D) $[\text{Fe}(\text{CN})_6]^{4-} > \text{PO}_4^{3-} > \text{SO}_4^{2-} > \text{Cl}^-$

Correct Answer: D. $[\text{Fe}(\text{CN})_6]^{4-} > \text{PO}_4^{3-} > \text{SO}_4^{2-} > \text{Cl}^-$

Solution:

The flocculating power of ions depends on their charge according to the Schulze-Hardy rule. Higher charged ions have greater flocculating power:

$$\text{Flocculating power} \propto \text{Charge on the ion.}$$

💡 Quick Tip

The Schulze-Hardy rule states that the coagulating power of an ion increases with its charge.

45. Gold sol is not a:

- (A) Macromolecular colloid

- (B) Lyophobic colloid
- (C) Multimolecular colloid
- (D) Negatively charged colloid

Correct Answer: (A) Macromolecular colloid

Solution:

Gold sol is a multimolecular colloid formed by the aggregation of gold atoms into clusters. It is lyophobic and negatively charged but not a macromolecular colloid, which consists of large molecules such as proteins or polymers.

💡 Quick Tip

Gold sol is an example of a multimolecular colloid, not a macromolecular colloid.

46. The incorrect statement about the Hall-Héroult process is:

- (A) Carbon anode is oxidised to CO and CO₂.
- (B) Na₃AlF₆ helps to decrease the melting point of the electrolyte.
- (C) CaF₂ helps to increase the conductivity of the electrolyte.
- (D) Oxidation state of oxygen changes in the overall cell reaction.

Correct Answer: D. Oxidation state of oxygen changes in the overall cell reaction.

Solution:

In the Hall-Héroult process, the oxidation state of oxygen does not change in the overall cell reaction. The process involves the reduction of alumina to aluminum and oxidation of carbon to CO and CO₂.

💡 Quick Tip

The Hall-Héroult process involves the reduction of aluminum but does not involve a change in the oxidation state of oxygen.

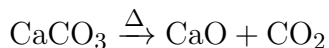
47. Select the correct statement:

- (A) Roasting involves heating the ore in the absence of air.
- (B) Calcination involves heating the ore above its melting point.
- (C) Smelting involves heating the ore with a suitable reducing agent and flux below its melting point.
- (D) Calcination of calcium carbonate is endothermic.

Correct Answer: D. Calcination of calcium carbonate is endothermic.

Solution:

Calcination is the process of heating an ore in the absence of air or in limited supply of oxygen. The decomposition of calcium carbonate during calcination is an endothermic process:

**💡 Quick Tip**

Calcination typically involves endothermic reactions, including the decomposition of carbonates and hydroxides.

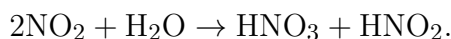
48. NO₂ gas is:

- (A) Colourless, neutral
- (B) Colourless, acidic
- (C) Brown, acidic
- (D) Brown, neutral

Correct Answer: C. Brown, acidic

Solution:

NO₂ is a brown-coloured gas and acidic in nature. It forms nitric acid when dissolved in water:

**💡 Quick Tip**

The acidic nature of NO₂ arises from its ability to form acids in aqueous solutions.

49. Identify the incorrect statement from the following:

- (A) Oxides of nitrogen in the atmosphere can cause depletion of the ozone layer.
- (B) Ozone absorbs the intense ultraviolet radiation of the Sun.
- (C) Depletion of the ozone layer is because of its chemical reactions with chlorofluoroalkanes.
- (D) Ozone absorbs infrared radiation.

Correct Answer: D. Ozone absorbs infrared radiation.

Solution:

Ozone primarily absorbs ultraviolet (UV) radiation and prevents it from reaching the Earth's surface. It does not absorb infrared radiation, which is absorbed by greenhouse gases like CO₂ and CH₄.

💡 Quick Tip

Ozone absorption is limited to UV radiation and does not contribute to the greenhouse effect.

50. The correct decreasing order of boiling points of hydrogen halides is:

- (A) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
- (B) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
- (C) $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
- (D) $\text{HI} > \text{HF} > \text{HBr} > \text{HCl}$

Correct Answer: C $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$

Solution:

The boiling point of HF is the highest due to strong hydrogen bonding. Among the remaining halides, the boiling point increases with molecular weight, following the order:

$\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$

💡 Quick Tip

Hydrogen bonding and molecular weight both influence the boiling points of hydrogen halides.

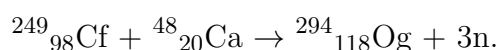
51. The synthetically produced radioactive noble gas by the collision of $^{249}_{98}\text{Cf}$ with $^{48}_{20}\text{Ca}$ is:

- (A) Radon
- (B) Radium
- (C) Oganesson
- (D) Xenon

Correct Answer: C. Oganesson

Solution:

The synthesis of Oganesson ($Z = 118$) involves the collision:



This produces Oganesson, the heaviest known noble gas.

💡 Quick Tip

Oganesson ($Z = 118$) is a synthetic element and the last member of the noble gas group in the periodic table.

52. The transition element ($\approx 5\%$) present with lanthanoid metal in Misch metal is:

- (A) Mg
- (B) Fe
- (C) Zn

(D) Co

Correct Answer: B. Fe

Solution:

Misch metal is an alloy composed of approximately 95% lanthanoid metals and 5% iron (Fe). It is widely used in lighter flints and other industrial applications.

💡 Quick Tip

Misch metal contains lanthanoids as the major component and iron as a minor component for added strength.

53. Match the following:

I. Zn^{2+}	i. d^8 configuration
II. Cu^{2+}	ii. Colourless
III. Ni^{2+}	iii. $\mu = 1.73 \text{ BM}$

Codes:

(A) i, ii, iii

(B) ii, iii, i

(C) ii, i, iii

(D) i, iii, ii

Correct Answer: B. ii, iii, i

Solution:

- Zn^{2+} has no unpaired electrons, making it colourless.
- Cu^{2+} has a magnetic moment of 1.73 BM due to one unpaired electron.
- Ni^{2+} has a d^8 configuration.

💡 Quick Tip

The colour and magnetic moment of ions depend on their electronic configuration and the number of unpaired electrons.

54. Which of the following statements related to lanthanoids is incorrect?

(A) Lanthanoids are silvery white soft metals.

(B) Samarium shows +2 oxidation state.

(C) Ce^{4+} solutions are widely used as oxidising agents in titrimetric analysis.

(D) Colour of lanthanoid ions in solution is due to d-d transitions.

Correct Answer: D. Colour of lanthanoid ions in solution is due to d-d transitions.

Solution:

The colour of lanthanoid ions in solution is due to f-f transitions, not d-d transitions. Lanthanoids have partially filled f-orbitals, and their electronic transitions occur within the f-orbital.

💡 Quick Tip

Lanthanoid ions exhibit colour due to f-f transitions, unlike transition metals which show d-d transitions.

55. A metalloid is:

- (A) Bi
- (B) Sb
- (C) P
- (D) Se

Correct Answer: B. Sb and 4. Se

Solution:

According to the NCERT classification of group 15 and 16 elements, antimony (Sb) and selenium (Se) are metalloids due to their intermediate properties between metals and non-metals.

💡 Quick Tip

Metalloids exhibit properties of both metals and non-metals and are typically semiconductors.

56. A pair of isoelectronic species having a bond order of one is:

- (A) N_2 , CO
- (B) N_2 , NO^+
- (C) O_2^- , F_2
- (D) CO, NO^+

Correct Answer: C. O_2^- , F_2

Solution:

O_2^- and F_2 are isoelectronic species with 18 electrons each. Their molecular orbital configurations result in a bond order of one.

💡 Quick Tip

Isoelectronic species have the same number of electrons and similar molecular properties such as bond order.

57. Identify the wrong relation for real gases:

(A) $Z = \frac{V_{\text{real}}}{V_{\text{ideal}}}$

(B) $p_{\text{real}} = p_{\text{ideal}} + \frac{an^2}{V^2}$

(C) $V_{\text{real}} = V_{\text{ideal}} - nb$

(D) $(p + \frac{a}{V^2})(V - b) = RT$

Correct Answer: A. $Z = \frac{V_{\text{real}}}{V_{\text{ideal}}}$

Solution:

The correct expression for the compressibility factor (Z) is:

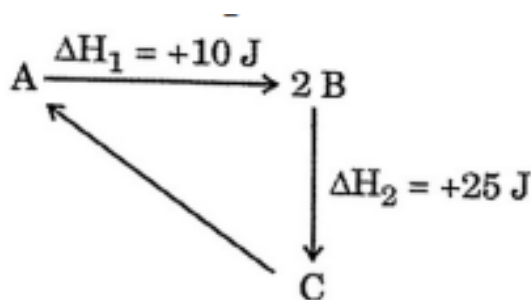
$$Z = \frac{p_{\text{real}}V_{\text{real}}}{RT}.$$

The relation $Z = \frac{V_{\text{real}}}{V_{\text{ideal}}}$ is incorrect because it does not account for deviations from ideality caused by intermolecular forces and molecular volume.

💡 Quick Tip

The compressibility factor (Z) measures how much a real gas deviates from ideal gas behavior.

58. From the diagram, $\Delta_r H$ for the reaction $C \rightarrow A$ is:



(A) +35 J

(B) -15 J

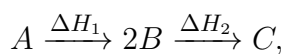
(C) -35 J

(D) +15 J

Correct Answer: C. -35 J

Solution:

Given the reaction:



$\Delta H_1 + \Delta H_2 = \Delta H(C \rightarrow A) = -35 \text{ J}$. Therefore, the enthalpy change for the reverse reaction ($C \rightarrow A$) is:

$$\Delta H = -35 \text{ J}.$$

💡 Quick Tip

Reverse enthalpy changes are equal in magnitude but opposite in sign to forward enthalpy changes.

59. For which one of the following mixtures is composition uniform throughout?

- (A) Sand and water
- (B) Grains and pulses with stone
- (C) Mixture of oil and water
- (D) Dilute aqueous solution of sugar

Correct Answer: D. Dilute aqueous solution of sugar

Solution:

A dilute aqueous solution of sugar is a homogeneous mixture where the sugar molecules are uniformly distributed throughout the solvent. In contrast, the other options represent heterogeneous mixtures with non-uniform compositions.

💡 Quick Tip

Homogeneous mixtures have uniform composition and properties throughout the solution.

60. The energy associated with the first orbit of He^+ is:

- (A) 0 J
- (B) -8.72×10^{-18} J
- (C) -4.58×10^{-18} J
- (D) -0.545×10^{-18} J

Correct Answer: B. -8.72×10^{-18} J

Solution:

The energy associated with an orbit is given by:

$$E_n = -2.18 \times 10^{-18} \cdot \frac{Z^2}{n^2} \text{ J.}$$

For He^+ , $Z = 2$ and $n = 1$:

$$E_1 = -2.18 \times 10^{-18} \cdot \frac{2^2}{1^2} = -8.72 \times 10^{-18} \text{ J.}$$

💡 Quick Tip

The energy of an electron in a hydrogen-like atom depends on the nuclear charge (Z) and the principal quantum number (n).